

edx Phot1x report

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Abstract—This report presents the design, simulation, fabrication, and experimental characterization of silicon-on-insulator (SOI) Mach–Zehnder interferometers (MZIs) developed as part of the UBCx Phot1x course. A set of unbalanced MZIs with different path length differences was designed to investigate the relationship between the free spectral range (FSR) and the waveguide group index. Waveguide simulations were performed using Lumerical MODE to determine the effective and group indices and to evaluate fabrication tolerances by means of a corner analysis. The fabricated devices were subsequently characterized experimentally, and the group index was extracted from the measured transmission spectra using both an autocorrelation-based method and an analytical transfer-function fit. The experimentally extracted group index agrees with the simulated value within 0.5%, demonstrating excellent agreement between simulation and experiment.

I. INTRODUCTION

A fundamental building block in integrated optics is the Mach–Zehnder interferometer (MZI), which exploits optical interference between two optical paths to realize wavelength filters, optical modulators, and phase-sensitive sensors. Integrated photonics enables the realization of these interferometric devices on a compact silicon-on-insulator (SOI) platform, making them attractive for applications in optical communications, sensing, signal processing, and optical computing.

An important parameter governing wave propagation in integrated waveguides is the group index, which determines the propagation velocity of optical pulses and directly influences the spectral response of interferometric devices. Accurate knowledge of the group index is therefore essential for both the design and characterization of silicon photonic circuits. A convenient method for extracting the group index is based on the transmission spectrum of an unbalanced Mach–Zehnder interferometer, whose free spectral range (FSR) is directly related to the waveguide group index and the optical path length difference between its two arms.

In this work, a set of unbalanced silicon Mach–Zehnder interferometers was designed on a silicon-on-insulator platform. The waveguide height was fixed at 220 nm and the nominal waveguide width was chosen as 500 nm, such that the waveguide operates in the fundamental quasi-TE mode around the telecom wavelength of 1550 nm. Six interferometers with optical path length differences of 30 μm , 60 μm , 100 μm , 150 μm , 250 μm and 500 μm were designed. This range of optical path length differences was selected to produce free spectral ranges between approximately 1.1 nm and 19 nm, ensuring that multiple interference fringes are observable within the measurement bandwidth while maintaining sufficient spectral resolution for accurate group-index extraction.

The objective of this work is to model, simulate, fabricate, and experimentally characterize these interferometers. By

comparing simulated and measured transmission spectra, the waveguide group index is extracted and validated using two independent analysis methods.

II. THEORY

A. Mach-Zehnder Interferometer

The interferometer consists of two optical paths that are split and recombined. The output intensity is determined by the phase difference of the two arms. The propagation constant of light traveling thorough the waveguide is

$$\beta = \frac{2\pi n_{\text{eff}}}{\lambda},$$

where n_{eff} is the effective index of refraction and λ the wavelength of the light.

If the two arms of the MZI have lengths L_1 and L_2 and the path length difference is $\Delta L = L_2 - L_1$, we can derive the phase difference:

$$\Delta\phi = \beta\Delta L = \frac{2\pi n_{\text{eff}}\Delta L}{\lambda}$$

If a single electric field E_{in} is split and recombined, the two split fields interfere on the beam splitter/combiner. The field at the output is then:

$$E_{\text{out}} = \frac{E_{\text{in}}e^{\phi} + E_{\text{in}}}{2}$$

and the resulting output intensity - also called the transfer function - is

$$I_{\text{out}} = T(\lambda) = \frac{1}{2}I_{\text{in}} (1 + \cos(\Delta\phi)).$$

If the combining beam splitter has two output ports, the second output port exhibits an additional phase shift of $\pi/2$.

B. Free Spectral Range (FSR)

We see constructive interference (in the case of two outputs, the other port shows destructive interference) when the phase difference is a multiple of 2π :

$$\Delta\phi = 2\pi m, \quad m \in \mathbb{Z}.$$

For two adjacent maxima at wavelengths (λ) and $(\lambda + \Delta\lambda)$, the phase condition leads to the definition of the free spectral range (FSR) [1].

$$\Delta\lambda = \frac{\lambda^2}{n_g\Delta L}$$

where (n_g) is the group index of the waveguide.

C. Group index

The group index that was introduced in the last equation is defined as:

$$n_g = n_{\text{eff}} - \lambda \frac{\partial n_{\text{eff}}}{\partial \lambda}$$

The group index accounts for dispersion and describes how the phase velocity changes with wavelength. It determines the spacing of the interference fringes in the MZI spectrum. We can rearrange the expression for the FSR:

$$n_g = \frac{\lambda^2}{\Delta \lambda \delta L}.$$

Therefore, by measuring the FSR ($\Delta \lambda$) from the transmission spectrum and the path length difference (ΔL) defined by the layout design, the group index can be determined.

III. MODELLING AND SIMULATION

A. Waveguide Geometry and Mode Analysis

The MZI is based on a standard silicon-on-insulator (SOI) strip waveguide, which is designed for the wavelengths in the telecom C-band (1550 nm). The geometry is optimized to support quasi-single-mode propagation in the quasi-TE polarization. The design parameters are summarized as follows:

- Core material: Silicon (Si), refractive index $n \approx 3.47$ at 1550 nm
- Cladding: Silicon dioxide (SiO₂) for both, upper and lower cladding ($n \approx 1.44$)
- Waveguide width: 500 nm
- Waveguide height: 220 nm
- Polarization: Quasi-TE mode

The chosen geometry provides strong optical confinement within the silicon core while maintaining evanescent field decay into the surrounding oxide. It is designed for single-mode operation in the quasi-TE polarization over the wavelength range of interest. A simulation of the fundamental TE mode was done with Lumerical MODE Solutions and can be seen in Fig. 1.

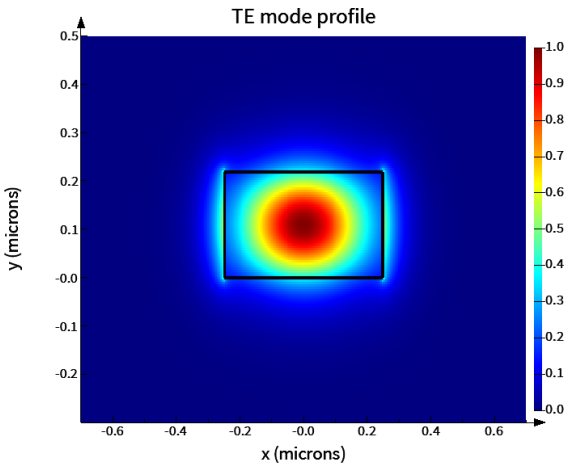


Figure 1. Simulated fundamental quasi-TE mode profile of the silicon strip waveguide with a height of 220 nm and a width of 500 nm.

B. Effective Index and Group Index

To get a better understanding of the dispersive properties of our designed waveguide, we computed the effective index (n_{eff}) and the group index (n_g) as a function of wavelength (See Fig.2,3). The simulations indicate a group index of ≈ 4.197 , which serves as a reference to validate the interferometric extraction method.

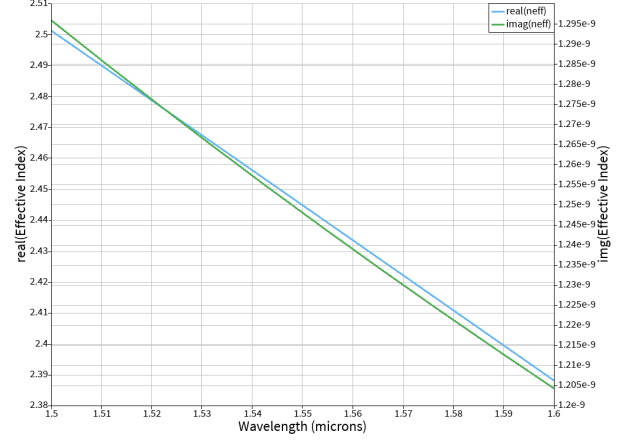


Figure 2. Simulated effective index of the 500 nm $\times$ 220 nm silicon strip waveguide over the telecom C-band.

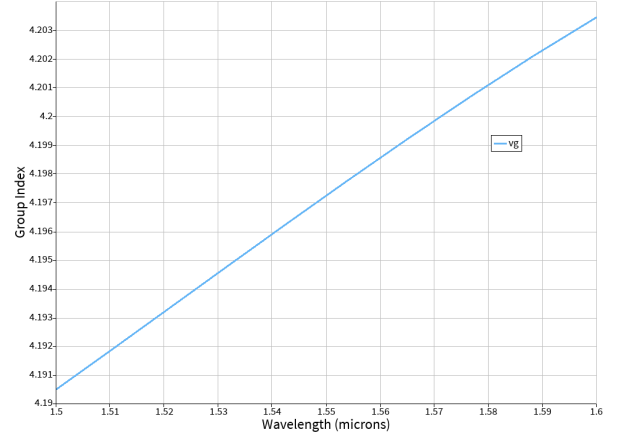


Figure 3. Simulated group index of the 500 nm $\times$ 220 nm silicon strip waveguide over the telecom C-band.

C. Compact Model of the Waveguide

For circuit-level simulation, we can model the wavelength-dependent effective index by fitting a second-order polynomial:

$$n_{\text{eff}}(\lambda) = a_0 + a_1(\lambda - \lambda_0) + a_2(\lambda - \lambda_0)^2.$$

By fitting our simulated wavelength sweep in Lumerical MODE, we can directly extract the coefficients. A second-order polynomial fit of the simulated effective index yields

$$n_{\text{eff}}(\lambda) = 2.447 - 1.133(\lambda - 1.55) + 0.044(\lambda - 1.55)^2.$$

D. Corner Analysis

To evaluate the sensitivity of the Mach-Zehnder interferometer to fabrication tolerances, a corner analysis was performed on the nominal silicon strip waveguide with a cross-section of $500 \text{ nm} \times 220 \text{ nm}$. The analysis considered expected process variations in both the waveguide width and the silicon thickness. The waveguide width was varied between 470 nm and 510 nm , while the silicon thickness was varied between 215.3 nm and 223.1 nm .

Five simulations were carried out using Lumerical MODE: one for the nominal geometry and one for each of the four corner combinations. For each configuration, a frequency sweep around 1550 nm was performed. The effective index was fitted as a function of wavelength, and the group index was extracted from the fitted dispersion.

The extracted group index was then used to calculate the expected free spectral range of the MZI with the exact path length difference $\Delta L = 250.252 \mu\text{m}$. The results are summarized in Table I and visualized in Fig. 4.

Table I
CORNER ANALYSIS OF THE SILICON STRIP WAVEGUIDE. THE RESULTING GROUP INDEX IS USED TO PREDICT THE FSR OF THE MZI WITH $\Delta L = 250.252 \mu\text{m}$.

Width [nm]	Thickness [nm]	n_g	FSR [nm]
470	215.3	4.24737	2.26030
470	223.1	4.25569	2.25588
500	220.0	4.19716	2.28734
510	215.3	4.17639	2.29871
510	223.1	4.18334	2.29490

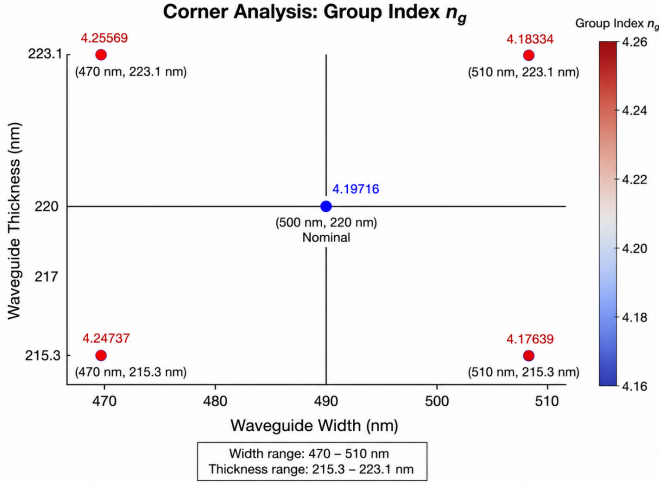


Figure 4. Corner-analysis results showing the simulated group index for the five waveguide geometries.

The nominal waveguide geometry yields a simulated group index of $n_g = 4.19716$, corresponding to an expected FSR of 2.28734 nm . Across the simulated fabrication corners, the group index varies from 4.17639 to 4.25569 , while the predicted FSR varies from 2.25588 nm to 2.29871 nm . This defines the expected performance window of the fabricated MZI due to waveguide width and thickness variations.

E. Layout Design

The complete photonic integrated circuit was designed using KLayout together with the SiEPIC PDK. The layout contains six unbalanced Mach-Zehnder interferometers with optical path length differences of $30 \mu\text{m}$, $60 \mu\text{m}$, $100 \mu\text{m}$, $150 \mu\text{m}$, $250 \mu\text{m}$, and $500 \mu\text{m}$. Each interferometer consists of two 3-dB directional couplers connected by silicon strip waveguides with a nominal cross-section of $500 \text{ nm} \times 220 \text{ nm}$.

Surface grating couplers are placed at the chip edge to couple light into and out of the photonic circuit using a fiber array. The layout also includes alignment structures and waveguide routing required for fabrication and optical characterization.

Figure 5 shows the final mask layout of the fabricated photonic integrated circuit.

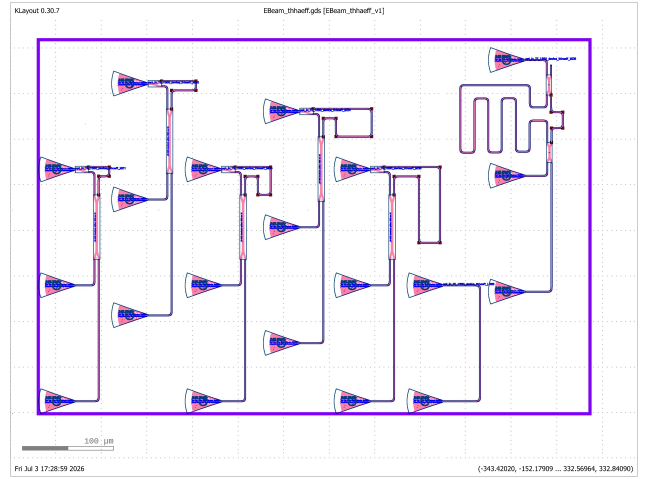


Figure 5. KLayout mask layout of the fabricated silicon photonic circuit containing six Mach-Zehnder interferometers with different optical path length differences. The devices share common input and output grating couplers for optical characterization.

F. Design verification and extraction of group index

The relationship between the optical path length difference of an MZI and its free spectral range allows the waveguide group index to be determined experimentally. Before characterizing the fabricated devices, the expected spectral response of each interferometer can be predicted using the simulated group index obtained from the waveguide analysis. These simulated values serve as a reference for the subsequent comparison with the experimental measurements.

In addition, the complete MZI was simulated using Lumerical INTERCONNECT. Figure 6 shows the simulated transmission spectrum of the interferometer with a path length difference of $100 \mu\text{m}$. The complementary output spectra exhibit the expected interference pattern together with the wavelength-dependent response of the grating couplers.

Table II summarizes the calculated and simulated free spectral ranges for all designed interferometers.

The chosen values for the path difference are in Table 1, together with the expected and simulated group index values.

Table II
CALCULATED FREE SPECTRAL RANGE FROM THE SIMULATED GROUP INDEX COMPARED WITH THE SIMULATED MZI RESPONSE OBTAINED FROM LUMERICAL INTERCONNECT FOR THE DESIGNED INTERFEROMETERS.

ΔL [μm]	Calculated FSR [nm]	INTERCONNECT FSR [nm]
30	19.08	18.71
60	9.54	9.50
100	5.72	5.75
150	3.82	3.85
250	2.29	2.30
500	1.14	1.14

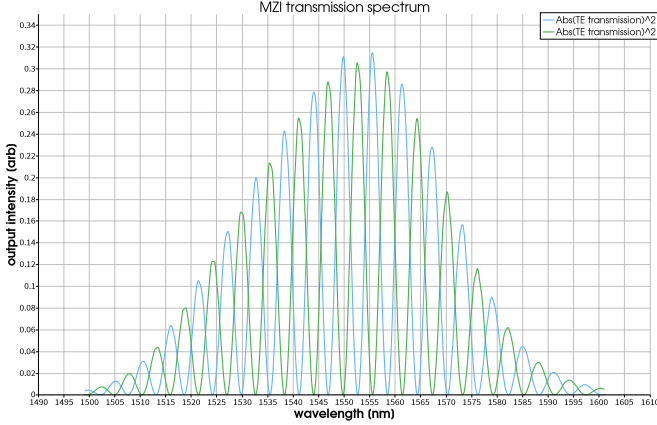


Figure 6. Simulated transmission spectra of both output ports of the Mach-Zehnder interferometer with an optical path length difference of 100 μm obtained using Lumerical INTERCONNECT.

IV. FABRICATION

A. Applied Nanotools, Inc. NanoSOI process:

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. [2](<http://www.appliednt.com/nanosoi>; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 μm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 $\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2$) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a Raith EBPG 5000+ electron beam instrument using a raster step size of 5 nm. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process

using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 μm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

V. EXPERIMENTAL DATA

The fabricated devices were characterized using the automated silicon photonics probe station available in the SiEPIC laboratory [3]. The setup consists of an Agilent 81600B tunable laser serving as the optical source and Agilent 81635A optical power sensors for transmission measurements. Light was coupled into and out of the chip using polarization-maintaining fiber arrays and surface grating couplers [4].

For all measurements presented in this work, the laser wavelength was swept from 1500 nm to 1600 nm with a wavelength step size of 10 pm. The input polarization was adjusted to excite the fundamental TE mode of the silicon strip waveguides.

Figure 7 shows the measured transmission spectra at the output ports of the fabricated Mach-Zehnder interferometer with an arm length difference of $\Delta L = 250 \mu\text{m}$. The periodic interference fringes resulting from the optical path length difference are clearly visible. In addition to the interference pattern, the spectrum exhibits a slowly varying envelope caused by the wavelength-dependent coupling efficiency of the grating couplers. The removal of this envelope and the subsequent extraction of the group index are described in the following section.

VI. EXPERIMENTAL DATA ANALYSIS

The objective of the experimental analysis is to extract the group index of the fabricated silicon waveguide from the measured transmission spectrum of the Mach-Zehnder interferometer (MZI). For this purpose, the MZI with an arm length difference of $\Delta L = 250 \mu\text{m}$ was selected, since its expected free spectral range (FSR) of approximately 2.29 nm provides a sufficient number of interference fringes within the measured wavelength range while maintaining good spectral resolution.

The analysis consists of four consecutive steps:

- 1) Removal of the grating coupler envelope,
- 2) Determination of the average FSR using autocorrelation,
- 3) Fitting the analytical MZI transfer function,
- 4) Extraction of the waveguide group index.

A. Removal of the Grating Coupler Envelope

The measured transmission consists of the periodic interference pattern generated by the MZI, superimposed on the wavelength-dependent coupling efficiency of the input and output grating couplers. This slowly varying envelope obscures

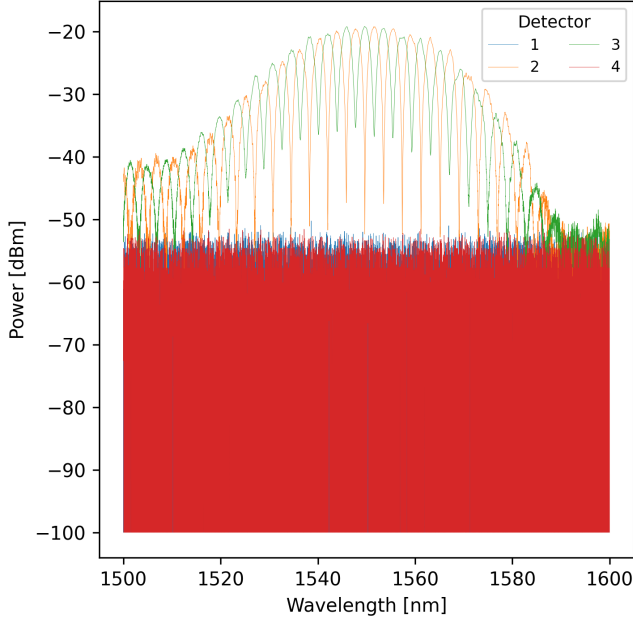


Abbildung 7. Measured transmission spectra at the output ports of the fabricated Mach-Zehnder interferometer with an arm length difference of $\Delta L = 250 \mu\text{m}$. The periodic interference fringes are superimposed on the wavelength-dependent response of the grating couplers.

the interference fringes and therefore has to be removed before further analysis.

To estimate the grating coupler response, a ninth-order polynomial was fitted to the measured transmission spectrum. Since the grating coupler efficiency varies only slowly with wavelength, the polynomial accurately represents the envelope while leaving the high-frequency interference fringes unaffected.

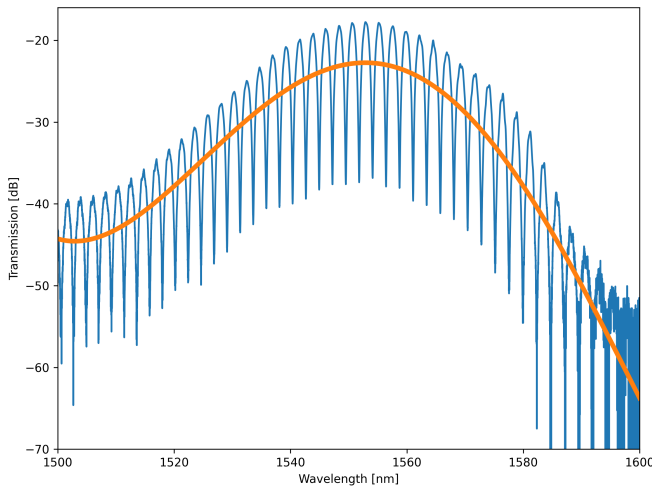


Abbildung 8. Measured transmission spectrum together with the ninth-order polynomial used to estimate the grating coupler envelope.

The measured transmission was subsequently normalized by the fitted polynomial, resulting in a corrected transmission spectrum with a nearly constant baseline. Only the wavelength

interval from 1530 nm to 1570 nm was retained for further processing, eliminating the low-transmission regions near the edges of the grating coupler bandwidth.

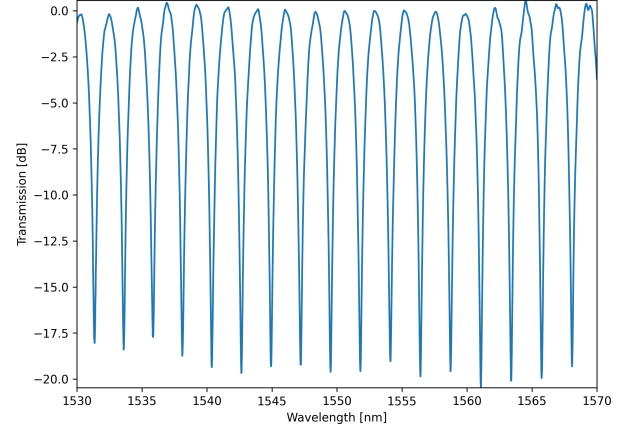


Abbildung 9. Corrected transmission spectrum of one output port after envelope removal and restriction to the wavelength interval from 1530 nm to 1570 nm.

B. Determination of the Free Spectral Range

The corrected transmission spectrum exhibits a highly periodic interference pattern, allowing the free spectral range to be determined. Rather than measuring the spacing between individual transmission peaks, an autocorrelation-based approach was employed. The autocorrelation emphasizes the dominant periodicity of the interference fringes while reducing the influence of measurement noise and local amplitude variations.

The autocorrelation of the corrected spectrum was calculated, and only the positive-lag portion was considered. The first significant maximum after the zero-lag peak corresponds to one period of the interference fringes and therefore to the free spectral range of the interferometer.

The measured FSR is obtained from

$$\text{FSR} = N_{\text{peak}} \cdot \langle \Delta \lambda \rangle,$$

where N_{peak} is the lag position of the first autocorrelation maximum and $\langle \Delta \lambda \rangle$ is the average wavelength spacing of the measurement.

Using the extracted lag position, the free spectral range was found to be

$$\text{FSR}_{\text{AC}} = 2.296 \text{ nm}.$$

With the exact optical path length difference of

$$\Delta L = 250.252 \mu\text{m},$$

which was extracted directly from the layout using the SiEPIC tools in KLayout, and the operating wavelength of

$$\lambda = 1550 \text{ nm},$$

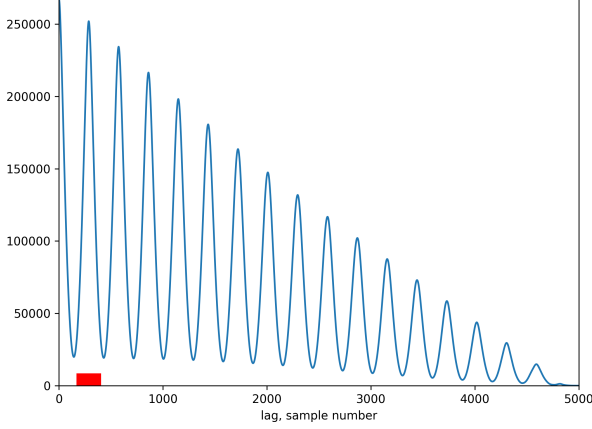


Abbildung 10. Autocorrelation of the corrected transmission spectrum. The first positive-lag maximum is used to determine the free spectral range of the interferometer.

the average group index was calculated as

$$n_{g,AC} = \frac{\lambda^2}{\Delta L \cdot \text{FSR}} = 4.1813.$$

The autocorrelation therefore provides a robust estimate of the average group index of the fabricated waveguide.

C. Transfer Function Fitting

While the autocorrelation provides an accurate estimate of the average FSR, it does not account for the wavelength dependence of the effective index. To obtain a more complete description of the measured spectrum, the corrected transmission data was fitted directly using the analytical transfer function of the Mach-Zehnder interferometer.

The wavelength dependence of the effective index was approximated by a second-order Taylor expansion around the operating wavelength,

$$n_{\text{eff}}(\lambda) = a_0 + a_1(\lambda - \lambda_0) + a_2(\lambda - \lambda_0)^2,$$

with $\lambda_0 = 1550$ nm.

Substituting this approximation into the analytical MZI transfer function results in a compact model that simultaneously describes the interference pattern and the waveguide dispersion. The polynomial coefficients were determined by performing a nonlinear least-squares fit to the corrected transmission spectrum.

Figure 11 compares the measured transmission with the fitted transfer function. The resulting coefficient of determination,

$$R^2 = 0.99659,$$

demonstrates an excellent agreement between the analytical model and the experimental data.

From the fitted polynomial, the wavelength-dependent group index was calculated. At the operating wavelength of 1550 nm, the fitted model yields

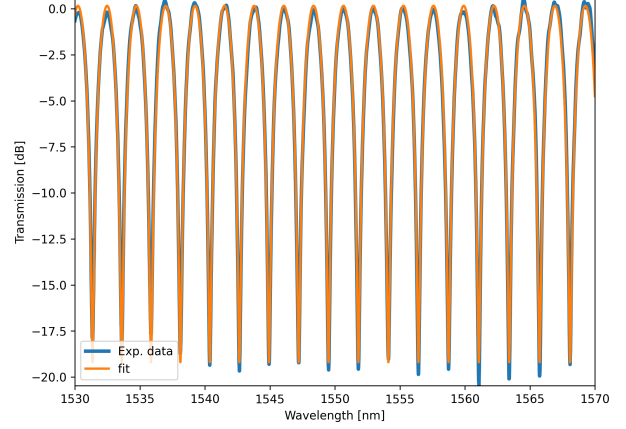


Abbildung 11. Corrected transmission spectrum together with the fitted analytical transfer function of the Mach-Zehnder interferometer.

$$n_g(1550 \text{ nm}) = 4.179289.$$

Using this fitted group index together with the extracted arm length difference results in a predicted free spectral range of

$$\text{FSR}_{\text{fit}} = 2.29712 \text{ nm}.$$

The excellent agreement between the fitted transfer function and the measured transmission demonstrates that the second-order Taylor expansion accurately captures the waveguide dispersion over the investigated wavelength range. Furthermore, the fitted group index agrees closely with the value obtained from the autocorrelation analysis, providing independent validation of the extraction method.

VII. COMPARISON OF SIMULATION AND EXPERIMENTAL RESULTS

The waveguide simulations performed in Lumerical MODE predicted a nominal group index of

$$n_g^{\text{sim}} = 4.197,$$

for the nominal waveguide geometry of $500 \text{ nm} \times 220 \text{ nm}$. Using the exact optical path length difference extracted from the layout,

$$\Delta L = 250.252 \text{ } \mu\text{m},$$

the expected free spectral range at the operating wavelength of $\lambda = 1550$ nm was calculated as

$$\text{FSR}_{\text{sim}} = 2.287 \text{ nm}.$$

Experimentally, two independent methods were used to determine the waveguide group index. The first employed an autocorrelation analysis of the measured transmission spectrum, while the second consisted of fitting the analytical transfer function of the Mach-Zehnder interferometer directly to the corrected measurement data. Table III summarizes the results obtained from the simulations and both experimental analysis methods.

Tabelle III
COMPARISON OF SIMULATED AND EXPERIMENTALLY EXTRACTED
WAVEGUIDE PARAMETERS.

Method	Group index n_g	FSR [nm]
Lumerical MODE simulation	4.1970	2.2870
Autocorrelation analysis	4.1813	2.2960
Transfer-function fit	4.1793	2.2971

The experimentally extracted group indices agree closely with the simulated value. The autocorrelation method yields a group index that differs from the simulation by approximately 0.37%, while the transfer-function fit results in a deviation of approximately 0.42%. Likewise, the experimentally determined free spectral ranges differ from the simulated prediction by less than 0.5%.

These small deviations are well within the expected variations caused by fabrication tolerances, measurement uncertainty, and simplifications in the analytical transfer-function model. In particular, the corner analysis performed in Section III demonstrated that moderate variations in waveguide width and silicon thickness produce measurable changes in both the effective and group indices. Consequently, the measured values are consistent with the expected fabrication-induced deviations from the nominal design.

Overall, the excellent agreement between simulation and experiment validates both the numerical waveguide model and the analytical data analysis procedure. The close correspondence between the independently extracted group indices demonstrates that the combination of autocorrelation analysis and transfer-function fitting provides a reliable method for characterizing integrated Mach-Zehnder interferometers and determining the waveguide group index from experimental transmission measurements.

VIII. ACKNOWLEDGEMENTS

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